Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Parameter	Details
Date	19 October 2026
Team ID	
Project Name	Health360: Clinical & Healthcare Risk Analysis
Maximum Marks	4 Marks

1. Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
1	Data Access & Integration	Seamlessly ingest and integrate 5,000 structured clinical patient records containing demographic profiles (6 age tiers: Teenager to Senior), physiological vitals (Systolic BP, Fatigue presence), and lifestyle indicators (Physical activity levels, Alcohol, Smoking habits) from	High

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
		`smart_healthcare_dataset.csv`.	
2	Data Processing	Perform data cleansing, validation, and standardizing of numeric metrics; categorize age groups; normalize clinical risk scoring index (baseline mean 100.0); compute discrete aggregates for fatigue status (50.8% incidence) and exercise tiers (Level 0, 1, 2).	High
3	Business Analysis	Compute essential epidemiological KPIs and risk metrics: average systolic blood pressure across age cohorts (~134.4 mmHg), relative risk variance across exercise levels, cohort-specific fatigue rates, and multi-variable correlations between substance use and cardiovascular stress.	High
4	Business Insights & Analytical Outputs	Provide 6 diagnostic visualization assets: (1) Average BP Across Age Bar Chart, (2) Demographic Pie Chart, (3) Fatigue Distribution Stacked Bar, (4) Exercise	High

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
		Level Donut Chart, (5) Alcohol & Smoking Consumption Bar Chart, (6) 6×3 Exercise vs Age Risk Heatmap, alongside a structured 5-Point Clinical Data Story.	
5	User Interaction & Analysis	Deliver multi-dimensional interactive slice-and-dice capabilities: demographic age filtering, exercise tier drill-downs, fatigue condition toggles, and dynamic KPI metric cards allowing clinicians to explore high-risk sub-populations.	Medium
6	Data Update & Reporting	Support periodic data refreshes for clinical cohorts, export comprehensive diagnostic risk summaries in visual and tabular formats, and provide automated preventive risk alerts for clinical decision support.	Medium

2. Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive, responsive web UI (deployed on GitHub Pages) and Tableau interface featuring clean typography, clear visual color codes for clinical risk states, and zero-learning-curve navigation for medical officers and analysts.
NFR-2	Security	Strict compliance with healthcare data privacy; de-identification and anonymization of patient records; encrypted data transmission over HTTPS; restricted administrative access.
NFR-3	Reliability	99.9% uptime for the web analytical portal; consistent chart rendering without data distortion or calculation drift across Chart.js and Tableau Public engines.
NFR-4	Performance	Fast dashboard page load time (< 2.5 seconds); instantaneous chart filtering and interactive cross-filtering (< 500 ms response latency) across all 5,000 patient records.

NFR No.	Non-Functional Requirement	Description
NFR-5	Availability	24/7 web-based accessibility via standard cloud platforms (GitHub Pages and Tableau Public Cloud) accessible simultaneously across desktop workstations and tablet devices.
NFR-6	Scalability	System architecture capable of scaling smoothly from 5,000 patient records to 100,000+ longitudinal records without architecture re-design or degradation in visualization rendering speed.